CROSSFIT GAMES ATHLETES-2019

1. Project Overview and Objective

This project focuses on analyzing a competitive sports dataset containing information about participants, their performances. The goal of the project is to clean, process, and analyze the data to extract meaningful insights about competitor performance, ranking patterns, and scoring distribution. The dataset also includes some inconsistencies making data preprocessing an essential part of the analysis.

---

2. Data Sources

- Source Description and Timeline: Data government/mavenanalytics.io/ Google Dataset Search.
- Domain: Sports

---

3. Problem Statement

- To transform **unstructured and inconsistent data** into a **clean, unified dataset** that can be used to evaluate athlete performance across events and identify top performers and rankings
- To analyze factors affecting performance and to compare performances between male and female athletes.
- Analyze consistency across events among countries.

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Height, weight	numeric	The details of a person
Name	String	Name of the person
Age	Numeric	Age of the person
Category	String	The category to which he comes under
Compitetor _id	Numeric	The id of each player
Rank	Numeric	The rank of each players
ordinal	Integer	Sequential identifier used to order records uniquely.
heat	Integer	Heat or round number in which the competitor participated.
points	Integer	Score or points earned by the competitor.
Breakdown	Text	Category or grouping variable used for segmentation or analysis.

5. Tools & Technologies

- Excel: Data cleaning, transformation.
- Power BI: Data modelling, visualization, and interactive dashboard creation.

---

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

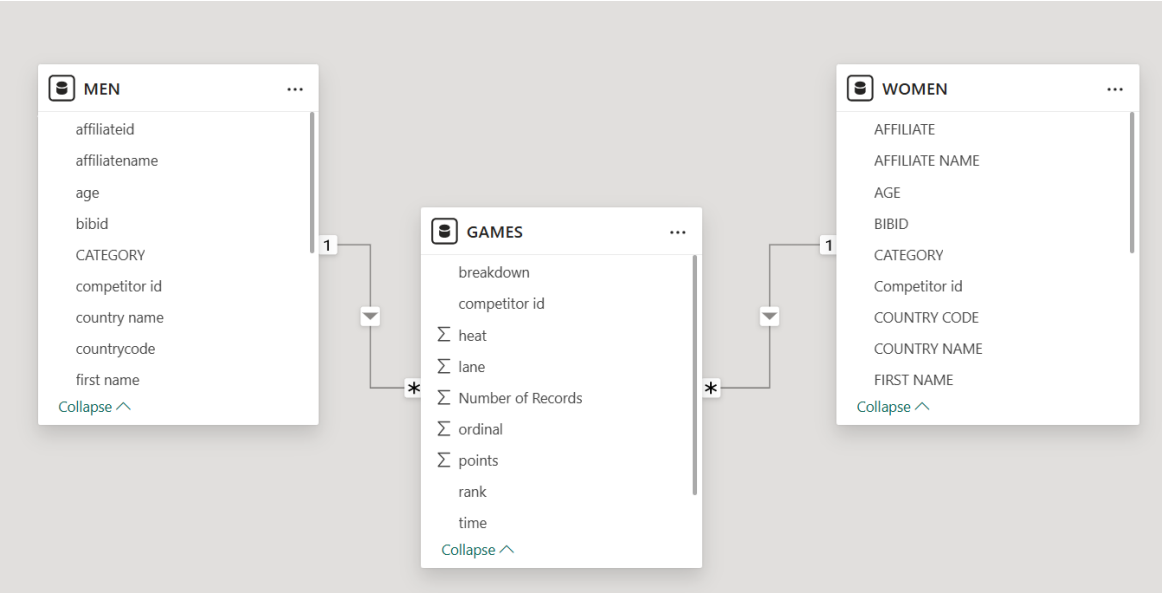
- Data Cleaning & Transformation: Removed duplicates, handled missing values, standardized formats.
- Filtering & Sorting: Organized data to focus on relevant records.

---

7. Data Modeling

- Data Model: Established relationships between tables, defined cardinality, and created lookup tables where necessary.

SCREENSHOT OF DATA MODELLING

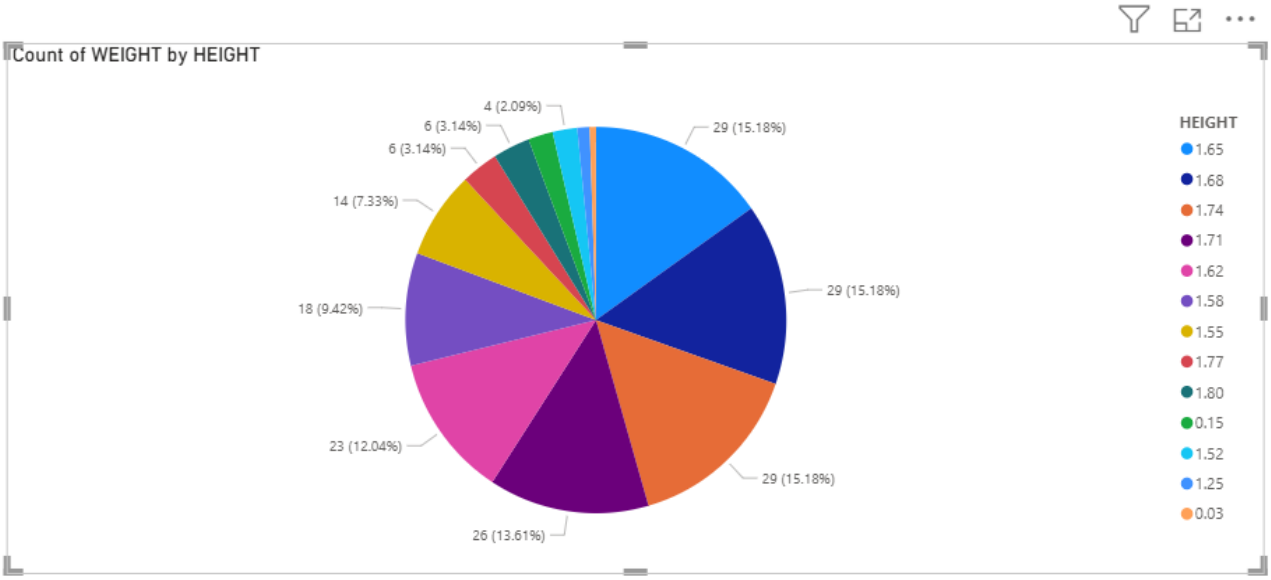


8.Analysis and Visualizations (Power BI)

Dashboard Features:

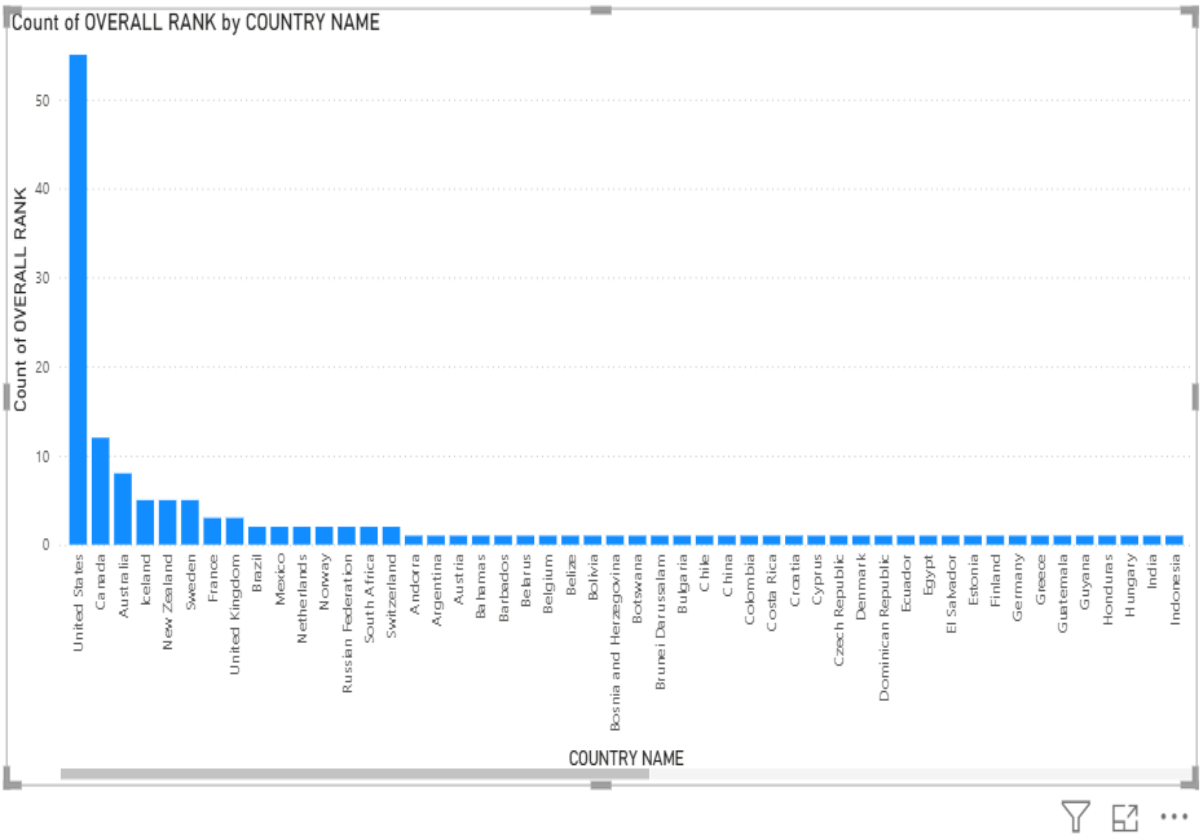
- Multiple Visualizations based on problem statement: Bar charts, line charts, pie charts, funnel, and tables to communicate insights.

SCREENSHOT OF DASHBOARD



Distribution of weight counts across different height categories

- Most individuals fall within a **moderate height range (1.65 m – 1.74 m)**, with fewer observations at extreme heights.
- The dominance of mid-range heights suggests the dataset reflects a **typical population structure rather than extreme variations**.
- If used for modeling, **height can be a strong predictor of weight trends**.



The bar chart displays the number of overall rankings recorded for each country.

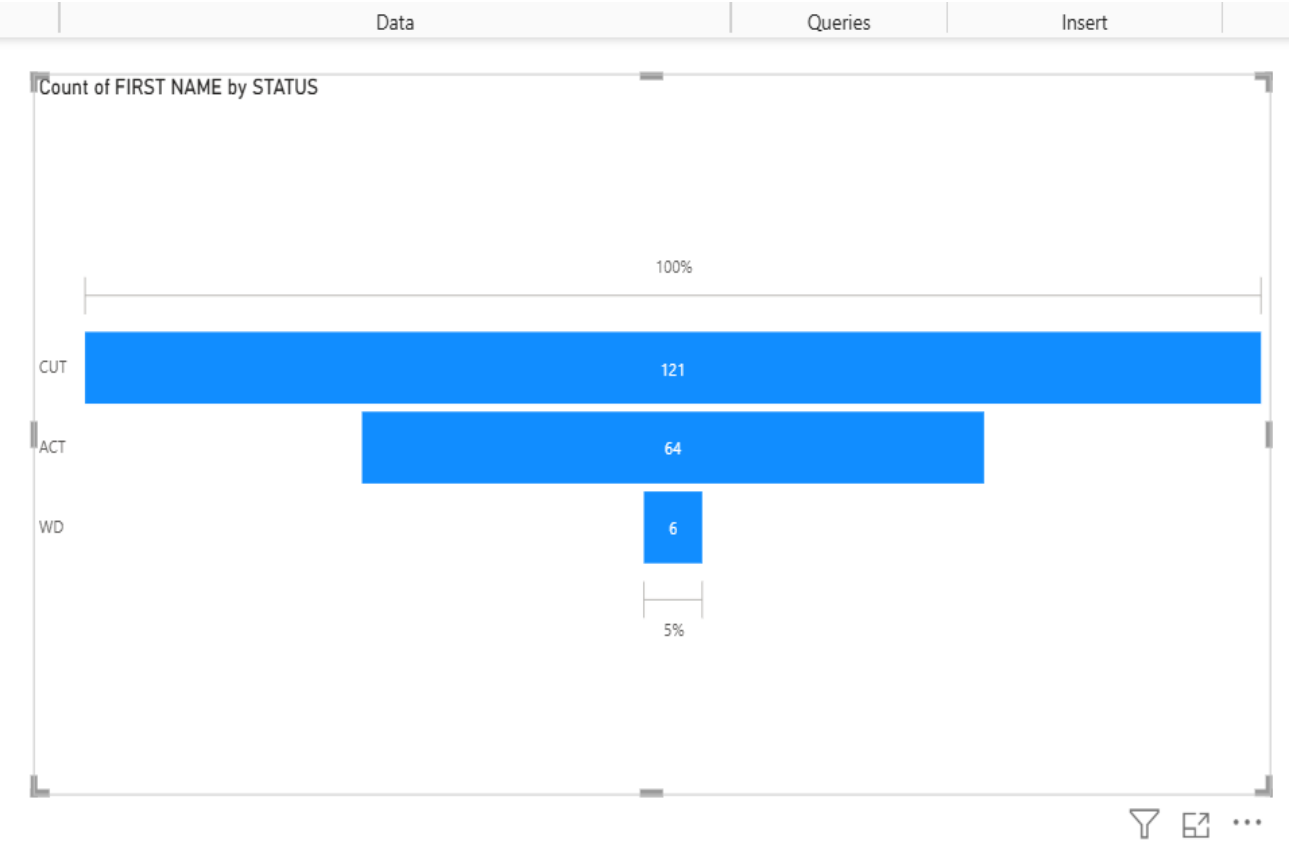
- The **United States** has the highest count, significantly exceeding all other countries .The next highest contributors include**:** Canada**,**Australia**,**Iceland. Most other countries have very low counts (1–3 records).

- Countries with more athletes or participants in the event naturally generate more ranking records. The dominance of a few countries suggests unequal participation or sampling, rather than a globally balanced dataset.
- Increasing global participation could reduce the gap between countries**.** High-representation countries may also produce more top-ranked performers.

country name	Count of first name	FIRST NAME
United States	68	
Canada	11	
Australia	5	
Iceland	4	
Italy	3	
Sweden	3	
United Kingdom	3	
Brazil	2	
Denmark	2	
France	2	
Poland	2	
South Africa	2	
Switzerland	2	
Afghanistan	1	
Algeria	1	
Andorra	1	
Angola	1	
Argentina	1	
Austria	1	
Bahamas	1	
Bahrain	1	
Barbados	1	
Belarus	1	
Belgium	1	
Total	**203**	

The table shows the number of participants from each country.

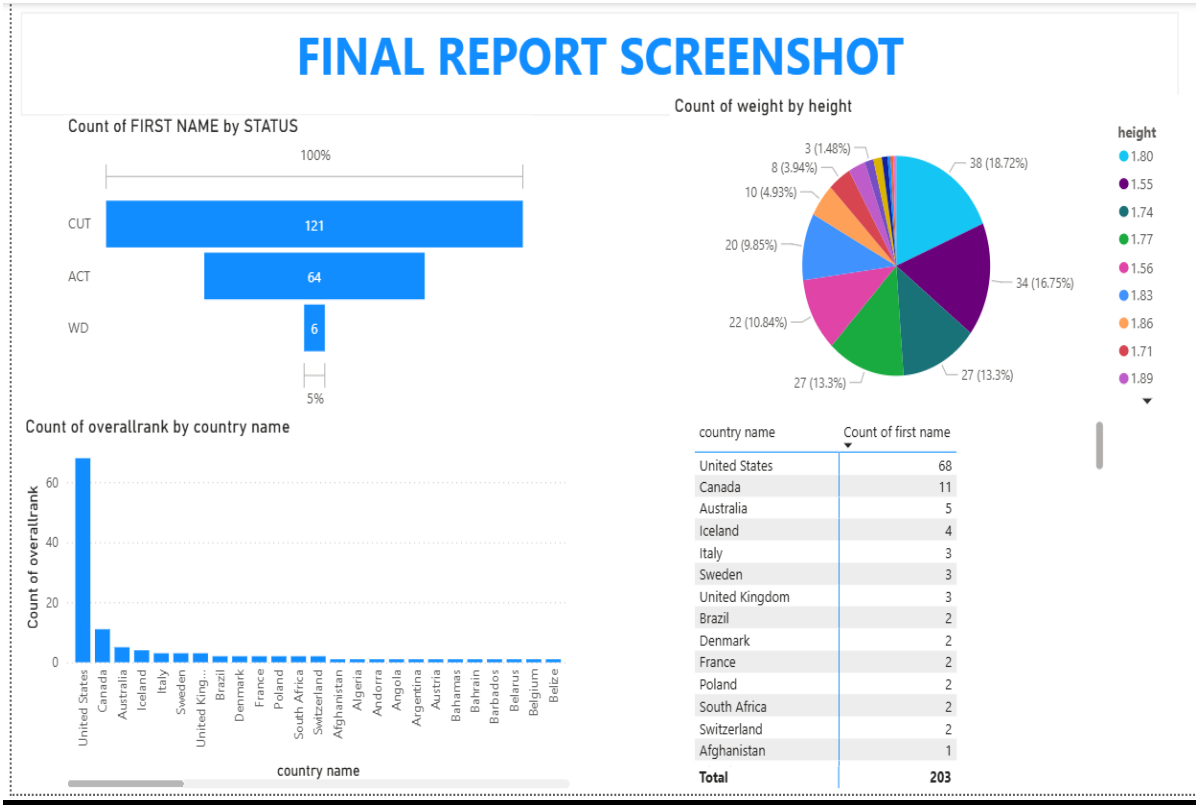
- The total number of participants is 203. A large portion of participants comes from a few dominant countries, while most countries contribute only a small number.
- Larger or more active countries in the event naturally have more participants. The dataset is not evenly distributed globally, indicating possible sampling bias or unequal participation.
- Increasing global outreach could lead to more balanced representation.
- High-participation countries may also produce more top-ranked individuals.



The chart shows the distribution of participants based on their status categories

- CUT has the highest count with 121 participants. ACT is the second largest group with 64 participants. WD (Withdrawn) has the lowest count with only 6 participants.
- The high number of CUT participants indicates a competitive filtering process, where many do not progress to the next stage. The dataset reflects a typical competitive environment where elimination is common.
- Most participants will continue to be eliminated at early stages . Only a smaller proportion will remain active (ACT).Withdrawal rates will likely stay minimal under normal conditions.

SCREENSHOT OF DASHBOARD /REPORT



INSIGHTS

High Proportion of CUT Status

The majority of records (around 60%) fall under the CUT category, indicating a strong filtering or rejection trend in the dataset.

Dominance of the United States

The United States contributes the highest number of records, accounting for about one-third of the total, showing a significant geographic imbalance.

Low Representation from Other Countries
Most other countries have very small counts (1–5 records), suggesting limited global diversity and potential sampling bias.

Ranking Influenced by Data Volume
Countries with higher participation, especially the United States, also show higher ranking counts, suggesting that rankings may be influenced by sample size rather than performance alone.

8. Conclusion

The analysis reveals significant global participation from both male and female athletes, emphasizing the inclusive nature of the competition. Higher-ranking participants consistently demonstrated superior performance metrics, notably in points and completion times. Physical attributes such as height and weight exhibited minimal influence on overall success. Instead, performance consistency and training affiliation emerged as key determinants of rank. The cleaned dataset thus provides a robust foundation for further analytical and comparative studies in competitive fitness.